Chamudi De Silva

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RESEARCH INTERESTS

Computational Cognitive Science, Computational Linguistics, LLM Evaluation, Pragmatics, and Human-Centered AI

EDUCATION

University of Peradeniya, Peradeniya, Sri Lanka 2021 – 2025
BSc. (Hons) in Computer Science, minor in Statistics
CGPA: 3.55/4.00 — Second Class Honours (Upper Division)
GPA for most recent 60 graded credits: 3.73/4.00

Selected coursework

- *Computer Science*: Artificial Intelligence, Neural Networks and Deep Learning, Machine Learning, Design and Analysis of Algorithms, Distributed Systems
- *Statistics*: Statistical Data Mining, Sampling Techniques, Probability Theory, Non-Parametric and Categorical Data Analysis

RESEARCH EXPERIENCE

Ongoing Projects Ongoing

- Physics-Informed Machine Learning for History-Dependent Hemodynamic Responses in fMRI
  Supervisors: Dr. Ruwan Ranaweera, Prof. Amalka Pinidiyaarachchi
- EEG Correlations of Creativity, Personality, and Vocational Interest
  Supervisors: Prof. Roshan Ragel, Dr. Ruwan Ranaweera, Dr. Erunika Dayaratna
- Toward Ethnically Equitable Emotional Intelligence in LLM-Based Agents
  (Extending the Shades of Sentiment study (accepted, ISBM 2026))
  Supervisor: Dr. Erunika Dayaratna

A Multi-Dimensional Study of Combined XAI Techniques for Regression Jun – Sep 2026
Supervisor: Dr. Erunika Dayaratna

- Evaluated single and combined XAI methods (SHAP, DiCE, Anchors, ProtoDash) on faithfulness, actionability, stability, and consistency.
- Found that well-chosen combinations improve several criteria, depending on model and aggregation strategy.
- Built a prototype that turns combined explanations into audience-tailored natural-language guidance.

Evaluating Emotional Fairness Across Ethnic Identities in LLMs Feb – Jun 2026
Supervisor: Dr. Erunika Dayaratna

- Audited seven LLMs on emotion-conditioned character profiles, using fairness metrics such as Demographic Parity Difference.
- Found skew toward Male and Global North personas and emotional and occupational stereotyping in every model.
- Showed that repeated prompting amplified bias by driving models toward near-deterministic outputs.

Integrating CL and RL for Personalised Response Generation in Companion Agents Mar – Dec 2025
Supervisor: Dr. Ruwan Nawarathna

- Proposed a hybrid continual and reinforcement learning framework combining user persona modelling, memory adaptation, and RL-based response selection.
- Built a companion agent as an evaluation platform to test personalised, emotion-aware responses.
- Showed improved user-specific adaptation and more emotionally aligned responses.

MANUSCRIPTS

Shades of Sentiment: Evaluating Emotional Fairness Across Ethnic Identities in LLMs (Accepted) 2026
W. Kekulandara, **C. De Silva**, E. Dayaratna.
5th World Conference on Information Systems for Business Management – ISBM

Is More Explanation Better? A Multi-Dimensional Study of Combined XAI Techniques for Regression (Under Review) 2026
C. De Silva, W. Kekulandara, E. Dayaratna.
8th IEEE International Conference on Advancements in Computing – ICAC

Hybrid Continual and Reinforcement Learning Framework for Adaptive Intelligent Companionship (In Prep) 2026
C. De Silva, R. Nawarathna

TEACHING EXPERIENCE

Teaching Assistant Jan 2026 – Present
Dept. of Statistics and Computer Science, Faculty of Science, University of Peradeniya

- Conduct weekly programming laboratories for 200+ undergraduate students.
- Teach introductory C and Python.
- Deliver substitute lectures in Operating Systems Concepts.
- Develop/evaluate instructional materials and assessments.

TECHNICAL SKILLS

Programming Languages	Python, R, SQL
ML & Data Analysis	PyTorch, scikit-learn, NumPy, Pandas, SciPy, Matplotlib
Databases & Tools	MySQL, VS Code, Google Colab, LaTeX

LEADERSHIP & CO-CURRICULAR ENGAGEMENT

Computer Society University of Peradeniya (CSUP) 2024 – 2025
Editor
Editorial oversight of student submissions and event documentation.

IEEE Computer Society Student Chapter, University of Peradeniya 2024 – 2025
Inaugural Assistant Secretary
Supported establishment and coordination of the university chapter; oversight of editorial operations.

Student Academic Advisory Committee (SAAC) 2022 – 2026
Organising Committee Member
Co-organisation of the workshop series and multi-day official orientation programmes for incoming students of Faculty of Science.

Leo Club of University of Peradeniya 2023 – 2024
Chief Editor
Oversight of all editorial operations; management of a trilingual editorial team by supervising group members.

LANGUAGES

English (fluent) · **Sinhala** (native) · **Tamil** (reading)

PERSONAL INTERESTS

Reading	Contemplative poetry (Mary Oliver), fiction
Languages	Learning Spanish and Korean
Other	Travelling